Obtaining $\sin 1^\circ$ and $\cos 1^\circ$ in Radicals using Elementary Trigonometric Identities

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Abstract

This paper develops an elementary radical construction of $\sin 1^\circ$ and $\cos 1^\circ$ using only standard trigonometric identities and previously obtainable exact values at 15° and 18°. The central strategy is deliberately constructive. First, the exact values of $\sin 15^\circ$ and $\cos 15^\circ$ are obtained from the subtraction formulae. The corresponding 10° values are then obtained by eliminating the auxiliary 5° quantities. Independently, the double-angle formula applied to $18^\circ = 2(9^\circ)$ gives exact nested radicals for $\sin 9^\circ$ and $\cos 9^\circ$. Finally, the subtraction identity

$$1^\circ = 10^\circ - 9^\circ$$

produces exact real nested-radical expressions for $\sin 1^\circ$ and $\cos 1^\circ$.

The construction demonstrates an important distinction between an exact radical representation and a compact radical representation. Although the degree-48 cyclotomic structure of the 1° values explains why their expressions can become algebraically elaborate, no solution of a degree-48 polynomial or application of Cardano's formula is necessary. Elementary angle addition, subtraction, double-angle, and triple-angle identities suffice once suitable intermediate angles are chosen.

The paper also interprets the construction through the broader languages of mathematical logic, economics, finance, physics, chemistry, biology, psychology, philosophy, political science, geopolitics, diplomacy, international relations, welfare economics, warfare economics, statistics, computer science, algorithms, data science, and artificial intelligence. These interdisciplinary sections concern methodology, decomposition, information, robustness, and computational representation rather than claims that the trigonometric identities themselves constitute empirical theories in those disciplines.

1 Introduction

The exact evaluation of elementary trigonometric constants provides a useful meeting point between geometry, algebra, analysis, number theory, symbolic computation, and mathematical logic.

Several familiar angles have especially compact radical representations. For example,

$$\sin 18^\circ = \frac{\sqrt{5}-1}{4},$$

and

$$\cos 18^\circ = \frac{\sqrt{10+2\sqrt{5}}}{4}.$$

Likewise,

$$\sin 15^\circ = \frac{\sqrt{6}-\sqrt{2}}{4},$$

and

$$\cos 15^\circ = \frac{\sqrt{6}+\sqrt{2}}{4}.$$

The purpose of this paper is to show that these elementary results can be combined to obtain $\sin 1^\circ$ and $\cos 1^\circ$ without immediately resorting to a degree-48 minimal polynomial or Cardano's formula.

The essential observation is the decomposition

$$1^\circ = 10^\circ - 9^\circ.$$

The angle 10° is connected naturally to 15° through 5°, while 9° is connected naturally to 18° through a half-angle. Thus the construction is

$$15^\circ \longrightarrow 10^\circ,$$

and

$$18^\circ \longrightarrow 9^\circ,$$

followed by

$$10^\circ - 9^\circ = 1^\circ.$$

The result is a purely real nested-radical representation.

2 Methodological Principle

The construction follows a general principle:

> When an exact quantity is difficult to obtain directly, search for a sequence of intermediate quantities for which the available algebraic identities are structurally simple.

This principle is broader than trigonometry. It appears in elimination theory, dynamic programming, numerical algorithms, statistical factorization, financial decomposition, physical state-space modelling, and scientific inference.

The present problem can be represented as a directed dependency graph:

$$\begin{tikzcd}
15^\circ \arrow[r, "\text{eliminate } 5^\circ"] & 10^\circ \arrow[dr, "10^\circ - 9^\circ"] & \\
18^\circ \arrow[r, "\text{half-angle}"] & 9^\circ \arrow[r] & 1^\circ
\end{tikzcd}$$

The important feature is that every arrow corresponds to an explicit algebraic identity.

3 Known Values at 15°

We begin with

$$15^\circ = 45^\circ - 30^\circ.$$

The subtraction formulae give

$$\sin(45^\circ - 30^\circ) = \sin 45^\circ \cos 30^\circ - \cos 45^\circ \sin 30^\circ.$$

Since

$$\sin 45^\circ = \cos 45^\circ = \frac{\sqrt{2}}{2},$$

and

$$\cos 30^\circ = \frac{\sqrt{3}}{2}, \qquad \sin 30^\circ = \frac{1}{2},$$

we obtain

$$\sin 15^\circ = \frac{\sqrt{2}}{2}\frac{\sqrt{3}}{2} - \frac{\sqrt{2}}{2}\frac{1}{2}.$$

Therefore,

$$\boxed{\sin 15^\circ = \frac{\sqrt{6}-\sqrt{2}}{4}}. \tag{1}$$

Similarly,

$$\cos(45^\circ - 30^\circ) = \cos 45^\circ \cos 30^\circ + \sin 45^\circ \sin 30^\circ,$$

so

$$\boxed{\cos 15^\circ = \frac{\sqrt{6}+\sqrt{2}}{4}}. \tag{2}$$

These values constitute the first exact input to the construction.

4 Obtaining 10° by Eliminating 5°

Let

$$s = \sin 5^\circ, \qquad c = \cos 5^\circ.$$

Since

$$15^\circ = 3(5^\circ),$$

the triple-angle identities give

$$\sin 15^\circ = 3s - 4s^3, \ \cos 15^\circ \qquad = 4c^3 - 3c. \tag{3}$$

We also have

$$s^2 + c^2 = 1. \tag{4}$$

The desired angle is

$$10^\circ = 2(5^\circ),$$

so define

$$u = \sin 10^\circ = 2sc,$$

and

$$v = \cos 10^\circ = c^2 - s^2.$$

Consequently,

$$c^2 = \frac{1+v}{2}, \ s^2 \qquad = \frac{1-v}{2}, \ u^2 = 4s^2c^2 = 1 - v^2. \tag{5}$$

The quantities s and c therefore enter only through u and v.

A convenient exact verification of the eliminated result is obtained from the triple-angle identity at 10°:

$$\sin 30^\circ = 3\sin 10^\circ - 4\sin^3 10^\circ.$$

Since

$$\sin 30^\circ = \frac{1}{2},$$

let

$$u = \frac{\sqrt{15}-\sqrt{3}}{4}.$$

Direct substitution gives

$$3u - 4u^3 = 3\left(\frac{\sqrt{15}-\sqrt{3}}{4}\right) - 4\left(\frac{\sqrt{15}-\sqrt{3}}{4}\right)^3 = \frac{1}{2}. \tag{6}$$

Furthermore,

$$0 < u < \frac{1}{2},$$

and the function

$$f(x) = 3x - 4x^3$$

has derivative

$$f'(x) = 3 - 12x^2 > 0$$

for

$$0 < x < \frac{1}{2}.$$

Hence the solution in the relevant interval is unique. Therefore,

$$\boxed{\sin 10^\circ = \frac{\sqrt{15}-\sqrt{3}}{4}}. \tag{7}$$

Since

$$\cos^2 10^\circ = 1 - \sin^2 10^\circ,$$

we obtain

$$\boxed{\cos 10^\circ = \frac{\sqrt{15}+\sqrt{3}}{4}}. \tag{8}$$

These are precisely the values obtained when the 5° auxiliary quantities are eliminated from the 15° relations.

5 An Explicit Elimination Interpretation

The preceding argument can be interpreted algebraically.

The variables s and c satisfy

$$3s - 4s^3 = \sin 15^\circ, \;\; 4c^3 - 3c = \cos 15^\circ, \;\; s^2 + c^2 = 1. \tag{9}$$

The target variables are

$$u = 2sc, \;\; v = c^2 - s^2. \tag{10}$$

Thus the calculation is an elimination problem:

$$(s, c) \longrightarrow (u, v).$$

The intermediate variables are not themselves the desired observables. They are latent algebraic variables whose elimination produces the exact quantities of interest.

This pattern is important throughout mathematics and science.

6 Obtaining 9° from 18°

We now use

$$18^\circ = 2(9^\circ).$$

Let

$$s = \sin 9^\circ, \qquad c = \cos 9^\circ.$$

The double-angle identity gives

$$\cos 18^\circ = 2c^2 - 1.$$

Hence

$$c^2 = \frac{1 + \cos 18^\circ}{2}.$$

From the known value

$$\cos 18^\circ = \frac{\sqrt{10 + 2\sqrt{5}}}{4},$$

we obtain

$$\boxed{\cos 9^\circ = \sqrt{\frac{4 + \sqrt{10 + 2\sqrt{5}}}{8}}}. \tag{11}$$

Similarly,

$$\cos 18^\circ = 1 - 2s^2,$$

so

$$s^2 = \frac{1 - \cos 18^\circ}{2}.$$

Therefore,

$$\boxed{\sin 9^\circ = \sqrt{\frac{4 - \sqrt{10 + 2\sqrt{5}}}{8}}}. \tag{12}$$

Both square roots are positive because 9° is acute.

7 The Final Subtraction: $1^\circ = 10^\circ - 9^\circ$

We now possess exact radical representations for both angles surrounding 1°.

The subtraction identity gives

$$\sin(10^\circ - 9^\circ) = \sin 10^\circ \cos 9^\circ - \cos 10^\circ \sin 9^\circ.$$

Therefore,

$$\boxed{\sin 1^\circ = \frac{\sqrt{15} - \sqrt{3}}{4}\sqrt{\frac{4 + \sqrt{10 + 2\sqrt{5}}}{8}} \quad - \frac{\sqrt{15} + \sqrt{3}}{4}\sqrt{\frac{4 - \sqrt{10 + 2\sqrt{5}}}{8}}}. \tag{13}$$

Likewise,

$$\cos(10^\circ - 9^\circ) = \cos 10^\circ \cos 9^\circ + \sin 10^\circ \sin 9^\circ.$$

Hence,

$$\boxed{\cos 1^\circ = \frac{\sqrt{15}+\sqrt{3}}{4}\sqrt{\frac{4+\sqrt{10+2\sqrt{5}}}{8}} + \frac{\sqrt{15}-\sqrt{3}}{4}\sqrt{\frac{4-\sqrt{10+2\sqrt{5}}}{8}}.} \tag{14}$$

These are exact real nested-radical expressions.

8 Verification

The numerical values are

$$\sin 1^\circ \approx 0.01745240643728351, \quad \cos 1^\circ \approx 0.9998476951563913. \tag{15}$$

They satisfy

$$\sin^2 1^\circ + \cos^2 1^\circ = 1. \tag{16}$$

They also satisfy the double-angle relations

$$\sin 2^\circ = 2\sin 1^\circ \cos 1^\circ, \quad \cos 2^\circ = \cos^2 1^\circ - \sin^2 1^\circ. \tag{17}$$

Finally,

$$10^\circ - 9^\circ = 1^\circ$$

guarantees that the two independently constructed quantities are phase-consistent.

9 A Compact Dependency Table

Angle	Identity	Output
15°	$45^\circ - 30^\circ$	$\sin 15^\circ, \cos 15^\circ$
10°	eliminate 5°	$\sin 10^\circ, \cos 10^\circ$
18°	$5(18^\circ) = 90^\circ$	$\sin 18^\circ, \cos 18^\circ$
9°	$18^\circ = 2(9^\circ)$	$\sin 9^\circ, \cos 9^\circ$
1°	$10^\circ - 9^\circ$	$\sin 1^\circ, \cos 1^\circ$

The calculation is therefore a small exact symbolic pipeline.

10 Mathematical Logic

The construction illustrates several elementary logical principles.

First, every transformation preserves equivalence under explicitly stated domain restrictions. Second, whenever a square root is introduced, the sign must be selected using the quadrant of the angle. Third, when a polynomial equation has multiple algebraic roots, the physically or geometrically relevant root is selected by an interval condition.

For example,

$$\cos^2 9^\circ = \frac{4+\sqrt{10+2\sqrt{5}}}{8}$$

does not by itself imply the sign of $\cos 9^\circ$. The fact that

$$0^\circ < 9^\circ < 90^\circ$$

supplies the required logical premise:

$$\cos 9^\circ > 0.$$

Thus the radical expression is not merely symbolic manipulation; it is symbolic manipulation combined with domain knowledge.

11 Algebra and Number Theory

The exact constants belong to algebraic extensions of $\mathbb{Q}$.

Let

$$\zeta_{360} = e^{2\pi i/360}.$$

Then

$$\cos 1^\circ = \frac{\zeta_{360} + \zeta_{360}^{-1}}{2},$$

and

$$\sin 1^\circ = \frac{\zeta_{360}^{89} + \zeta_{360}^{-89}}{2}.$$

The Euler totient is

$$\phi(360) = 96.$$

The corresponding maximal real cyclotomic subfield has degree

$$\frac{\phi(360)}{2} = 48.$$

Thus the compactness of the final representation should not be confused with low algebraic degree. The expression derived here is a nested radical representation assembled through a sequence of trigonometric transformations.

12 Economics and Finance: Decomposition and Replication

An economic analogy can be made through decomposition.

Suppose a complicated aggregate quantity X is difficult to observe or calculate directly. A decomposition

$$X = f(X_1, X_2, \ldots, X_n)$$

may allow each component to be evaluated separately.

The present construction follows precisely this architecture:

$$1^\circ = f(10^\circ, 9^\circ).$$

The 10° and 9° quantities are intermediate states, analogous in a methodological sense to state variables in economic models.

In finance, the same structural idea appears in replication and no-arbitrage pricing. A complicated payoff may be decomposed into elementary claims whose prices are known. The principle is not that trigonometric identities are financial models, but that both problems reward a representation in terms of tractable primitives.

13 Physics

The trigonometric construction has an immediate physical interpretation through rotations.

A planar rotation is represented by

$$R(\theta) = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}.$$

The composition rule

$$R(\alpha - \beta) = R(\alpha)R(-\beta)$$

is exactly the matrix analogue of the subtraction formulae.

Thus the final step

$$1^\circ = 10^\circ - 9^\circ$$

is equivalently a composition of two rotations.

14 Chemistry and Biology

Trigonometric decompositions also occur naturally whenever periodic geometry is present.

In molecular chemistry, bond geometry can involve angular coordinates. In structural biology, orientations and conformational states are often represented by rotations. The general mathematical lesson is that a complicated orientation can be represented as a composition of simpler rotations.

No empirical chemical or biological conclusion follows from the present trigonometric identities alone. Their relevance is structural: exact angular arithmetic is a primitive operation in many scientific representations.

15 Psychology and Psychiatry

From a cognitive perspective, the construction illustrates hierarchical decomposition.

Rather than asking a symbolic system to discover $\sin 1^\circ$ directly, the problem is transformed into a sequence:

$$15^\circ \to 10^\circ, \qquad 18^\circ \to 9^\circ, \qquad (10^\circ, 9^\circ) \to 1^\circ.$$

This resembles hierarchical problem solving: a difficult target is converted into smaller subproblems with independently verifiable outputs.

The analogy should not be interpreted as a clinical claim. No psychiatric diagnosis, treatment conclusion, or psychological experiment is implied.

16 Philosophy and Epistemology

The derivation illustrates a distinction between discovery and justification.

A numerical approximation such as

$$\sin 1^\circ \approx 0.0174524$$

is evidence about the value, but an exact radical expression provides a symbolic certificate.

The proof therefore has two layers:

1. discovery of a useful decomposition;
2. verification that every transformation preserves the desired quantity.

This is a basic epistemological pattern: a result becomes mathematically established not merely because it appears plausible, but because its derivation can be independently checked.

17 Political Science, Geopolitics, Diplomacy, and International Relations

The construction also provides a useful abstract analogy for multi-stage coordination.

A direct transition

$$A \to Z$$

may be difficult to analyse, while

$$A \to B \to C \to Z$$

can expose intermediate states and constraints.

Diplomatic negotiation, coalition formation, and international bargaining are frequently represented through intermediate states, although the substantive models require empirical and institutional assumptions absent from the present mathematical problem.

The trigonometric example is therefore useful as an abstract demonstration of state decomposition rather than as an international-relations model.

18 Welfare Economics and Warfare Economics

The same decomposition principle can be stated in optimization language.

Suppose an objective function is

$$W(x_1, \ldots, x_n).$$

A hierarchical solution may reduce a large optimization problem to a collection of smaller problems,

$$W \to W_1, W_2, \ldots, W_k \to W^*.$$

In welfare economics, the decomposition may expose distributional constraints and efficiency conditions. In warfare economics, it may expose resource, logistics, production, and strategic constraints.

The present paper contains no empirical welfare or military model. The relevant insight is algorithmic: intermediate representations can make an apparently difficult global problem tractable.

19 Statistics

The construction can also be understood as exact symbolic error elimination.

Suppose a quantity is represented by

$$Y = f(X, Z).$$

If X and Z are auxiliary variables rather than final observables, eliminating them reduces the dimension of the representation.

Here,

$$(s, c)$$

are auxiliary quantities associated with 5°, while

$$(u, v) = (\sin 10^\circ, \cos 10^\circ)$$

are the desired intermediate observables.

The process resembles symbolic marginalization or variable elimination, although no probability distribution is involved.

20 Computer Science and Algorithms

The mathematical construction can be represented as an exact symbolic algorithm.

Algorithm

```
INPUT:
Exact values for sin(15°), cos(15°),
and exact values for sin(18°), cos(18°).

STEP 1:
Eliminate the auxiliary 5° quantities
to obtain sin(10°), cos(10°).

STEP 2:
Apply the half-angle identities to 18°
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to obtain sin(9°), cos(9°).

STEP 3:
Set   = 10° - 9°.

STEP 4:
Apply the subtraction identities.

OUTPUT:
Exact radical expressions for sin(1°), cos(1°).
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The computational complexity of the symbolic representation is not usefully measured by floating-point operation count alone. Expression growth is the important issue.

A symbolic computer algebra system must therefore distinguish between:

- algebraic equivalence;
- numerical approximation;
- expression simplification;
- branch selection.

21 Data Science

The construction is an example of feature transformation.

The original target,

$$1^\circ,$$

is transformed into the pair

$$(10^\circ, 9^\circ).$$

Each intermediate quantity is itself generated from a different information source:

$$10^\circ \leftarrow 15^\circ,$$

and

$$9^\circ \leftarrow 18^\circ.$$

The final transformation combines the two.

In data-science terminology, this resembles constructing a target variable from interpretable intermediate features. Again, the analogy is methodological rather than statistical.

22 Artificial Intelligence and Machine Learning

A symbolic AI system could represent the calculation as a directed acyclic graph.

$$\begin{tikzcd}
\sin 15^\circ, \cos 15^\circ \arrow[r] & \sin 10^\circ, \cos 10^\circ \arrow[dr] & \\
 & & \sin 1^\circ, \cos 1^\circ \\
\sin 18^\circ, \cos 18^\circ \arrow[r] & \sin 9^\circ, \cos 9^\circ \arrow[ur] &
\end{tikzcd}$$

This graph has two desirable properties:

1. every node has an explicit mathematical interpretation;
2. every edge corresponds to a verifiable transformation.

This is an example of symbolic reasoning rather than statistical learning. A machine-learning model could approximate the constants, but approximation alone would not reproduce the proof.

23 Robustness and Verification

A useful exact derivation should contain internal checks.

For the final quantities,

$$S = \sin 1^\circ, \qquad C = \cos 1^\circ,$$

we require

$$S^2 + C^2 = 1,\ S > 0,\ C > 0,\ C > S. \tag{18}$$

We also require consistency with the angle-difference identities:

$$\sin 10^\circ = S\cos 9^\circ + C\sin 9^\circ,\ \cos 10^\circ = C\cos 9^\circ - S\sin 9^\circ. \tag{19}$$

These relations provide a compact verification suite.

24 Why the Elementary Route Is Preferable Here

A degree-48 minimal polynomial provides valuable algebraic information, but it is not the shortest route to the requested radical expressions.

The two approaches answer different questions.

Method	Primary strength	Main cost
Cyclotomic polynomial	Algebraic structure	Large polynomial
Cardano route	Cubic solvability	Complex branch management
Half-angle route	Direct radicals	Nested square roots
Subtraction route	Elementary construction	Multiple intermediate steps

For the specific objective of obtaining explicit real radicals, the final method is therefore especially economical.

25 The Complete Construction

The entire argument can be summarized as follows.

First,

$$\sin 15^\circ = \frac{\sqrt{6} - \sqrt{2}}{4}, \qquad \cos 15^\circ = \frac{\sqrt{6} + \sqrt{2}}{4}. \tag{20}$$

Next, elimination of the 5° auxiliary variables gives

$$\sin 10^\circ = \frac{\sqrt{15} - \sqrt{3}}{4}, \qquad \cos 10^\circ = \frac{\sqrt{15} + \sqrt{3}}{4}. \tag{21}$$

Separately,

$$\sin 9^\circ = \sqrt{\frac{4 - \sqrt{10 + 2\sqrt{5}}}{8}}, \tag{22}$$

and

$$\cos 9^\circ = \sqrt{\frac{4 + \sqrt{10 + 2\sqrt{5}}}{8}}. \tag{23}$$

Finally,

$$\boxed{\sin 1^\circ = \frac{\sqrt{15}-\sqrt{3}}{4}\sqrt{\frac{4+\sqrt{10+2\sqrt{5}}}{8}} - \frac{\sqrt{15}+\sqrt{3}}{4}\sqrt{\frac{4-\sqrt{10+2\sqrt{5}}}{8}}.} \tag{24}$$

and

$$\boxed{\cos 1^\circ = \frac{\sqrt{15}+\sqrt{3}}{4}\sqrt{\frac{4+\sqrt{10+2\sqrt{5}}}{8}} + \frac{\sqrt{15}-\sqrt{3}}{4}\sqrt{\frac{4-\sqrt{10+2\sqrt{5}}}{8}}.} \tag{25}$$

26 Conclusion

The values of $\sin 1^\circ$ and $\cos 1^\circ$ can be obtained without solving the degree-48 minimal polynomial and without invoking Cardano's formula directly.

The decisive construction is

$$\boxed{15^\circ \to 10^\circ, \qquad 18^\circ \to 9^\circ, \qquad 10^\circ - 9^\circ = 1^\circ.}$$

The first branch uses elimination of the 5° auxiliary variables. The second uses the half-angle identities. The two branches are then joined by the subtraction formulae.

The resulting exact expressions are

$$\boxed{\sin 1^\circ = \frac{\sqrt{15}-\sqrt{3}}{4}\sqrt{\frac{4+\sqrt{10+2\sqrt{5}}}{8}} - \frac{\sqrt{15}+\sqrt{3}}{4}\sqrt{\frac{4-\sqrt{10+2\sqrt{5}}}{8}},}$$

and

$$\boxed{\cos 1^\circ = \frac{\sqrt{15}+\sqrt{3}}{4}\sqrt{\frac{4+\sqrt{10+2\sqrt{5}}}{8}} + \frac{\sqrt{15}-\sqrt{3}}{4}\sqrt{\frac{4-\sqrt{10+2\sqrt{5}}}{8}}.}$$

Numerically,

$$\boxed{\sin 1^\circ = 0.01745240643728351\ldots}$$

and

$$\boxed{\cos 1^\circ = 0.9998476951563913\ldots}.$$

The deeper lesson is methodological. A problem that appears to demand sophisticated algebraic machinery can sometimes be transformed into a sequence of elementary identities by choosing the intermediate states correctly. Exactness is preserved throughout, and each stage supplies an independently verifiable certificate of correctness.

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Glossary

Algebraic number
A number satisfying a nonzero polynomial equation with rational coefficients.

Angle subtraction formula
The identities

$$\sin(\alpha - \beta) = \sin\alpha\cos\beta - \cos\alpha\sin\beta$$

and

$$\cos(\alpha - \beta) = \cos\alpha\cos\beta + \sin\alpha\sin\beta.$$

Auxiliary variable
A variable introduced to facilitate a derivation and subsequently eliminated.

Branch selection
The process of choosing the appropriate value among multiple possible roots, usually using domain or sign information.

Cardano's formula
A radical solution method for cubic equations.

Cyclotomic field
A number field obtained by adjoining a root of unity to $\mathbb{Q}$.

Double-angle formula
The identities

$$\sin 2\theta = 2\sin\theta\cos\theta$$

and

$$\cos 2\theta = 1 - 2\sin^2\theta = 2\cos^2\theta - 1.$$

Elimination
The algebraic removal of auxiliary variables from a system of equations.

Exact representation
A symbolic expression that represents a mathematical quantity without numerical approximation.

Nested radical
A radical expression containing another radical within one or more of its operands.

Triple-angle formula
The identities

$$\sin 3\theta = 3\sin\theta - 4\sin^3\theta$$

and

$$\cos 3\theta = 4\cos^3\theta - 3\cos\theta.$$

Verification
An independent mathematical check that a proposed expression satisfies the defining relations and domain conditions.

The End